

FIELD PROJECT — WASTE MANAGEMENT

SMART DUSTBIN

AI-Enabled Fill-Level Monitoring System

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Field Study Location	Goundanur, Coimbatore
Review Stage	Review 1 – 35% Project Completion

“Be smart, use smart dustbins.”

1. Abstract

Waste overflow is a practical waste-management challenge when collection is based on a fixed schedule rather than the actual condition of individual bins. The Smart Dustbin project proposes a low-cost monitoring approach that uses an ultrasonic sensor to estimate fill level and a camera with an object-detection model to visually confirm overflow or spillage. An ESP32 processes the sensing information, while Wi-Fi/GSM can communicate alerts to a dashboard or collection team. The proposed alert threshold is 80% capacity. This Review 1 report documents the project work completed toward the 35% milestone: field observation, problem identification, objectives, existing-solution analysis, proposed solution, system architecture, component selection, prototype concept and feasibility planning.

2. Introduction

The project was initiated to address the gap between actual bin fill conditions and fixed waste-collection schedules. During the field observation at Goundanur, Coimbatore, overflowing bins were observed by midday near the entrance, no wet/dry segregation was observed at the site, and the collection round occurred once a day before peak demand. These observations motivated the design of a smart monitoring system that can provide continuous visibility of bin status.

3. Field Study and Observations

- Field study location: Goundanur, Coimbatore.
- Bins were observed overflowing by midday near the entrance.
- No wet/dry waste segregation was observed at the site.
- One collection round was conducted daily, before peak demand.
- The identified gap was the absence of real-time visibility into the current fill level between collection rounds.

4. Problem Statement

Public dustbins can overflow before the next scheduled collection because collection is based on a fixed schedule rather than the actual fill level of the bin. This can result in waste spilling into surrounding areas, unhygienic conditions and delayed response to high-demand locations.

5. Objectives

- Detect the dustbin fill level in real time using an ultrasonic sensor.
- Generate an alert when the bin reaches the predefined 80% capacity threshold.
- Target a reduction in overflow incidents during the pilot period.
- Target a shorter response time from alert to waste pickup.
- Log fill-level data to identify peak-fill periods and support demand-based collection planning.

6. Existing System and Gap

Approach	Current Practice	Limitation
Fixed-schedule collection	Vehicles visit at set times	Ignores actual fill level and can over- or under-visit
Manual inspection	Staff physically check bins	Labour-heavy, infrequent and easy to miss
Public complaint reporting	Residents report overflow	Reactive; action occurs after the problem is visible

7. Proposed Solution

The shortlisted solution is a Smart Dustbin combining fill sensing and vision-based overflow detection. An ultrasonic sensor provides continuous fill-level information, while a camera and object-detection model provide visual confirmation of overflowing waste. The combination is intended to provide an additional confidence layer before an alert is acted upon.

Why this combination?

- The ultrasonic sensor provides continuous fill-level information.
- The sensor-based approach can continue to provide a reading even when visual conditions are not ideal.
- The camera and AI model can visually verify waste spillage or overflow.
- The combined approach is intended to reduce false alarms compared with relying on a single signal.

8. System Architecture

Stage	Component	Function
01 – Data Source	Ultrasonic sensor	Measures distance from the lid to the waste surface
02 – Processing	ESP32	Processes the reading and estimates fill percentage
03 – Communication	Wi-Fi / GSM	Transmits alert/status information
04 – Interface	Dashboard / SMS	Presents the condition to collection staff
Feedback	Servicing cycle	After collection, monitoring continues automatically

9. Hardware and Software Requirements

Category	Requirement	Purpose
Hardware	Smart dustbin	Waste collection container
Hardware	Ultrasonic sensor	Fill-level measurement
Hardware	ESP32	Processing and control

Category	Requirement	Purpose
Hardware	Camera	Image capture for overflow verification
Hardware	Wi-Fi/GSM	Communication and alerts
Software/AI	Arduino/ESP32 environment	Microcontroller programming
Software/AI	Object-detection model	Visual detection of bin/spillage
Software/AI	Dashboard / monitoring interface	Status display and operational response

10. Working Methodology

- Waste is deposited into the smart dustbin.
- The ultrasonic sensor measures the distance to the waste surface.
- The ESP32 processes the measurement and estimates the fill percentage.
- The estimated fill level is compared with the 80% threshold.
- The camera and AI model can provide visual confirmation of overflow/spillage.
- When the alert condition is reached, Wi-Fi/GSM can notify the collection team.
- The collection team services the bin.
- Monitoring continues after the bin is emptied.

11. AI / Object-Detection Component

The proposed AI component uses labelled images of overflowing bins. Bounding boxes can identify the bin and spillage regions so that an object-detection model can learn to recognize visible overflow. At the current 35% stage, the AI approach and training-data concept have been defined; model training and performance testing remain future work.

12. Prototype Status at 35%

The project has reached the initial prototype-design stage. The monitoring concept, hardware selection, AI approach, system architecture and alert logic have been defined. The physical hardware integration, model training and complete end-to-end testing are planned for the next development phase.

Activity	Status at Review 1
Field problem identification	Completed
Field observation	Completed
Problem statement	Completed
Objectives	Defined
Existing-system analysis	Completed
Proposed solution	Defined

Activity	Status at Review 1
Hardware selection	Defined
AI approach	Defined
System architecture	Designed
Prototype concept	Designed
Hardware integration	Future work
AI model training	Future work
End-to-end testing	Future work

13. Expected Outcomes

Parameter	Current Approach	Targeted Smart-Dustbin Approach
Collection trigger	Fixed schedule	Actual fill level at 80% threshold
Overflow incidents	Frequent / unpredictable	Target: reduce by half during pilot
Response time	Same day or later	Target: under 2 hours from alert
Visibility	None between rounds	Continuous fill-level data

The values in the outcome targets are project targets, not measured final results at this stage.

14. Feasibility and Risk Mitigation

- Technical feasibility: the proposed architecture uses a small set of commonly available sensing, processing and communication components.
- Operational feasibility: alerts can help collection staff prioritize bins according to actual condition.
- Economic feasibility: the project is intended as a low-cost prototype; exact unit cost remains to be finalized.
- Risk – sensor fouling: protective housing and a regular cleaning schedule are proposed.
- Risk – false alerts: sensor readings can be complemented by visual AI confirmation.

15. Future Work

- Assemble and integrate the hardware prototype.
- Calibrate the ultrasonic sensor and verify fill-level calculations.
- Program the ESP32 and implement threshold logic.
- Integrate the camera.
- Collect and label overflow images.
- Train and evaluate the object-detection model.
- Implement Wi-Fi/GSM notifications.

- Develop the monitoring dashboard.
- Conduct controlled prototype testing and field validation.

16. Conclusion

The Smart Dustbin project converts a field-observed waste-management problem into a measurable technical solution. The current 35% milestone establishes the problem, objectives, proposed sensing and AI approach, system architecture and prototype plan. The next stage will focus on physical integration, software implementation, AI training and testing. The long-term goal is to support cleaner public spaces by enabling collection teams to respond to actual bin conditions rather than relying only on fixed schedules.

17. AI Usage Declaration

AI assistance was used for research support, report/slide drafting and presentation or diagram layout. All project-specific content should be reviewed and edited by the student before submission to ensure that the declaration accurately reflects the assistance actually used.

End of Review 1 – 35% Project Report